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Image Forgery Detection using Deep Learning Model

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***Abstract*—**The act of tampering with or altering an image is known as image forgery. This image manipulation may appear to be harmless. When an image is used in legal proceedings or forensic investigations, however, it must be unaltered to maintain the integrity of the evidence. Manipulation of an image can be done in a variety of ways. There are numerous ways to predict image forgery, just as there are numerous ways to predict forgery. However, the problem with those methods is that they are so advanced that they are inaccessible to the general public. To resolve this issue, this study comes up with the idea of the developing of a Deep Learning (DL) model. For this purpose, a dataset consisting of both original and forged images is collected from a database named CASIA. This dataset is then preprocessed. The preprocessing techniques include image resizing and denoising. The processed images are then used to train the DL model. The working of the model is then tested using the accuracy and loss as the main parameters. In the end, it is found that the model is capable of producing an accuracy greater than 95% which is greater than many other DL models. Just like the accuracy, the loss value is also found to be very low that it can be neglected. This makes the model more efficient. In the future the model can be deployed into a website, making it useful for the common people.

Keywords— ***Image processing, preprocessing techniques, transfer learning, Forgery.***

I. INTRODUCTION

Image forgery is a process where the actual image is tampered with or modified. This manipulation of an image may seem unharmful. But when an image is used for legal actions or forensic investigations, it has to be unadulterated to maintain the truthfulness of the evidence. In most cases, the authenticity of the images plays a major role and it is investigated before a final decision. There are various ways in which one can manipulate an image.The image forgery includes the process of altering the pixels of the image, increasing or decreasing the saturation and brightness levels, and modifying the anomalies of an image. Just like the forgery, there are so many ways to predict the image forgery as well. But the problem with those methods is that they are so advanced that they cannot be used by common people. Because the fraudulent are so good at concealing the major factors that can reveal the fact that the media is forged, it is extremely difficult to predict false and forged images and videos these days. This problem can be solved by employing the model developed in this study. This is done using the DL model which was constructed using VGG-19 architecture. The construction and working of the model are explained clearly in the upcoming chapters.

II. LITERATURE SURVEY

The researcher uses image classification and semantic segmentation in the detection of image forgery [1]. This study states that the accurate detection of copy-move is

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vague when the ground truth is generated by the Convolutional Neural Network (CNN). Thus, an effective technique is required in the field of image forensics. This study uses different layers of segmentation and classification. Thus, it can provide greater accuracy and F1 score. As mentioned before, various technologies have been employed in the detection of image forgery. The journal [2] uses the CNN framework in the detection of image forgery.This study states that the CNN framework was used to increase the number of input images that can be fed into the model. It also states that the image resizing during analysis often reduces the working of the model. But the main disadvantage of this study is that it is so complicated to be used in daily life.Just like forgery detection, forgery localization is also an important branch of forgery detection that can precisely identify the area that has been forged in the image. The authors[3] proposed an idea of PRNU-based forgery localization. This study also uses image segmentation for this purpose. The term PRNU is an acronym for Photo-Response Non-Uniformity. The localization technique is based on the multi-orientation scheme. Due to this, this model can effectively identify the parts that are manipulated even if two or more parts of the image are modified. Another study [4] uses speeded-up robust feature (SURF) and polar complex exponential transform (PCET) methods in the field of image forgery detection. SURF expands to SURF and PCET is an acronym for Polar Complex Exponential Transform. The study detects forgery in four steps. In the first step, the images are divided into blocks using super-pixel segmentation and made sure there is no overlapping of pixels. In the next step, the main features of the images are extracted. The SURF and PCET are then used to eliminate the unwanted features from the image as the third step. As the final step, the model is morphed mathematically to ensure the same performance everytime.

The writer[5] who used semantic segmentation also proposed another idea in the field of image forensics. This study explains the usage of Texture analysis of median filter for detection of spliced images. The study states that image splicing is one of the major threats in the modern world as it can be done in such a way no one can notice it at the first glance. To detect such forgery, this study uses Median Filter Residual or the MFR. It is found that the method proposed can identify spliced images with an accuracy of over 90%.The median filter method was also used by the article[6]. This study develops a model that can detect whether the entered JPEG image is compressed or not. Sometimes, image forgery can be easily identified when the image was compressed during forgery. For this purpose, this study investigated median filters and trained them to detect image compression. In the end, the filter designed in this study is also able to produce results with higher accuracy. One of the major concerns in the process of image forgery detection is the poor quality of the images. To resolve this issue, the authors [7] have developed a median filter that is capable of detecting image forgery even with low-quality images. This is achieved by widening the filter layers of the input images. This way the quality of the image can be preserved before detection. This study uses a CNN for the classification of images as forged and original.Image sharpening is also a part of image forgery. When an image is sharpened, the image loses its original authenticity. Thus, it is mandatory to check whether an image is sharpened or not before a legal purpose. The experiment[8] has proposed a method to detect image sharpening. In this study, two different sets of pixels are generated from the image, and the difference between the two sets is analyzed. By doing so, the fact of whether the image is sharpened or not can be detected. The article [9] has also proposed the construction of a morphological filter detector that can be used in image forensics. This study also uses SVM as the main classifier and uses grayscale filtering for better results. Average accuracy of about 78% was achieved by this study. Another research by a group of researchers from the Nanjing University of China has proposed a modified CNN that can be used in image classification and forensics [10]. This CNN is known as the Quaternion CNN or the QCNN and is capable of detecting forgery from colored images.

III. MATERIALS AND METHODS

A dataset consisting of both original and forged images is obtained from a database named CASIA database. The images then undergo certain processes to train and test the efficiency of the DL model. The processes are depicted as a flowchart in figure 1.

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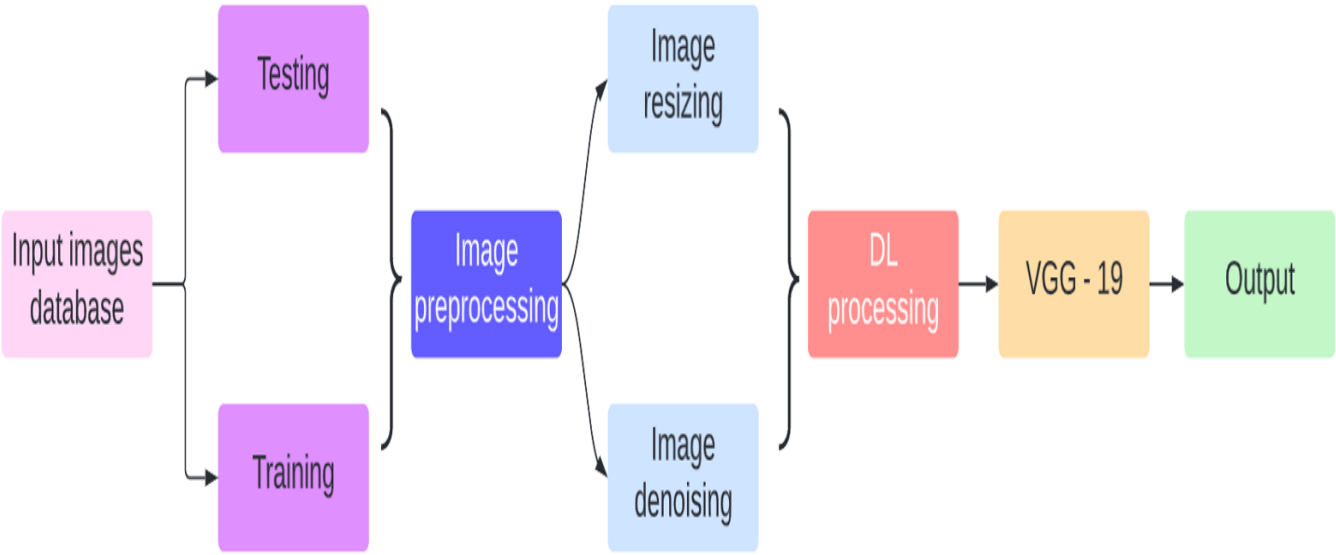


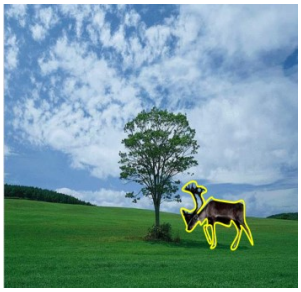
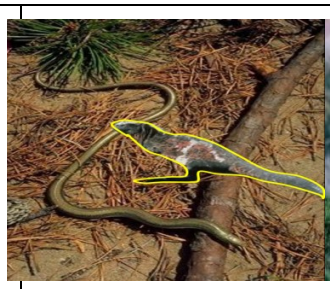
Figure 1. Workflow of the study

As shown in the figure, the images are first split into two parts for testing and training. The split images are preprocessed and are used to train and test the DL model. The efficiency of the model is then analyzed and recorded.

IV. Data Collection And Preprocessing

A dataset consisting of approximately 13000 images is obtained. This dataset is known as a CASIA database [11]. The database used in this study is the second version of the CASIA database and it contains about 13000 images with almost 7500 original images and 5000 forged images. The sample of the images is shown in table 1.

Table 1. Sample data

Sl. No	Forge	Original
1		
2		
3		

As mentioned inthe above table, the dataset consists of both original and forged images. The distribution ratio of the images between original and forged images is represented as a pie chart in figure 2.

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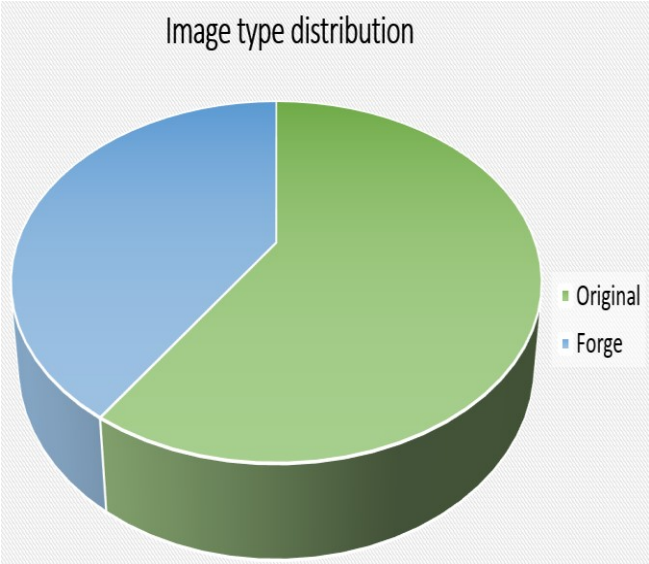

Figure 2. Image distribution

The images are used for both testing and training the model. For this purpose, the images have to be divided further. The images are split based on what is shown in table 2.

Table 2. Data split up

CASIA 2.0	Total	Train	Test
Original	7492	5993	1499
Forge	5124	4100	1024
Total	12616	10093	2523

From the table, it is clear that the major part of the image database is used to train the DL model. But before training the model, the images must be preprocessed to ensure efficient working of the model. The preprocessing techniques include image resizing and image denoising.

A. Image resizing

In computer vision, image resizing is a necessary preprocessing step. All computer vision models train faster on smaller photos in general. Furthermore, the images must be identical in several deep learning and machine learning version designs. Regardless, the unprocessed images can be of a variety of sizes, including exclusive sizes. As a result, all of the available photos were deleted and cropped to a specific size[12].

B. Image denoising

Sometimes, images obtained from the database consist of contamination caused due to the quality of the camera. Other causes of noise in images include the temperature of the environment which is immutable in certain cases. These contaminations are known as noise. The noise in the image can hinder the efficiency of the computer vision model. The process of removing or reducing the impact of the noise from the images is known as image denoising. This is one of the necessary preprocessing steps when computer vision uses deep learning.At times, researchers also use computer vision in the process of image denoising [13].

V. TRANSFER LEARNING

A computer-generated classification system for photos, texts, and even sounds is known as a deep learning model. This deep learning model can be created using a variety of deep learning methods. Among the techniques used are CNN, radial basis function networks, recurrent neural networks, transfer learning, and others. This study uses transfer learning in the detection of image forgery. The utilization of transfer learning increases the efficiency of the working of various computer vision models [14].

A. VGG -19

The term VGG – 19 is an acronym for Visual Geometry Group – 19. This is a CNNthat has 19 layers. This algorithm is an updated version of both AlexNet and VGG - 16 which was developed by the researchers of the Oxford University. Studies state that the average precision and accuracy level of this algorithm is higher than 90% in most cases [15]. Figure 3 shows the architecture of the VGG – 19 algorithms.

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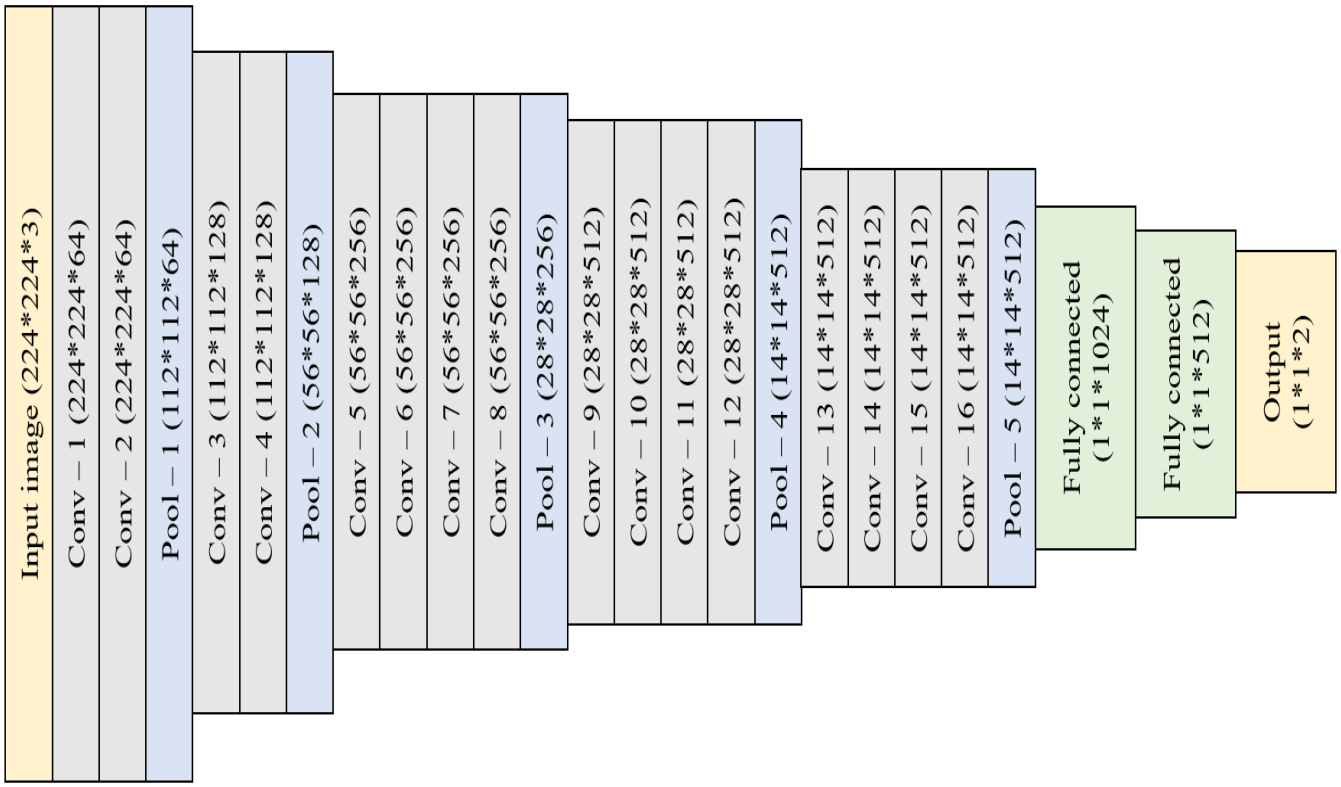

Figure 3. VGG –19 architecture

From the above figure, it can be inferred that the VGG – 19 algorithm consists of various layers. The input image is first sent to the pooling layers with certain dimensions which are known as the convolutional layers. The output of these layers will be the same image with lesser dimensions. The image then enters into another pooling layer of lower dimension. These layers are known as max-pooling layers. The output of the max-pooling layers is sent to the convolutional layer again. This process is repeated until the image reaches the smallest dimension possible. The images then reach the fully connected layers. The fully connected layers and the successive layers are used to collect the features that are required for the classification process. One of the major advantages of the VGG -19 architecture is that the hidden layers reduce the need to write codes for every process. The layers of the model are trained in such a way that they can detect the forgery of images. This training is done using the preprocessed images from the database.

VI. RESULTS AND DISCUSSION

A dataset consisting of original images and forged images is collected from a database named CASIA. This dataset consists of almost 12000 images out of which 7500 images are original and 5000 images are forged approximately. These images are then preprocessed to maintain uniform dimensions. The preprocessing techniques include image resizing and denoising. Simultaneously, a DL model is developed using a transfer learning algorithm. The construction of the model uses VGG–19 architecture. The model is then trained using the preprocessed images. The training is repeated for several epochs and the results of each epoch are recorded and tabulated. The results of the training are shown in table 3.

Table 3. Results of the training

Epoch	Accuracy	Loss
0	0.4521	0.6842
1	0.5479	0.6147
2	0.6679	0.5029
3	0.7886	0.4484
4	0.815	0.4098
5	0.84	0.3636
6	0.8596	0.3299
7	0.8599	0.3109
8	0.885	0.2654
9	0.8766	0.2751
10	0.9167	0.2194
11	0.9174	0.201
12	0.908	0.2175
13	0.9246	0.1875
14	0.931	0.1614
15	0.9419	0.1488
16	0.9528	0.1244
17	0.9515	0.1243
18	0.9524	0.1066

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19	0.9535	0.1067

From table 3, it can be seen that the accuracy of the model is quite low in the initial epochs. Also, the loss value is very high. But as the number of epochs increases, the accuracy increases and the loss value decreases. By the final epochs, the accuracy of the model is higher than 90 and the loss value is relatively very low than that of the initial epochs. During the 20th epoch i.e., the final epoch, the accuracy of the model is 0.9535 which is great for a DL algorithm that is used in the detection of image forgery. The loss value of the model is 0.1067 which is so low that it's neglectable.

As the results of the training are constructive, the model is then tested using the smaller part of the processed dataset. The results of the testing are also tabulated. The accuracy of the model during the testing is represented graphically in figure 4.

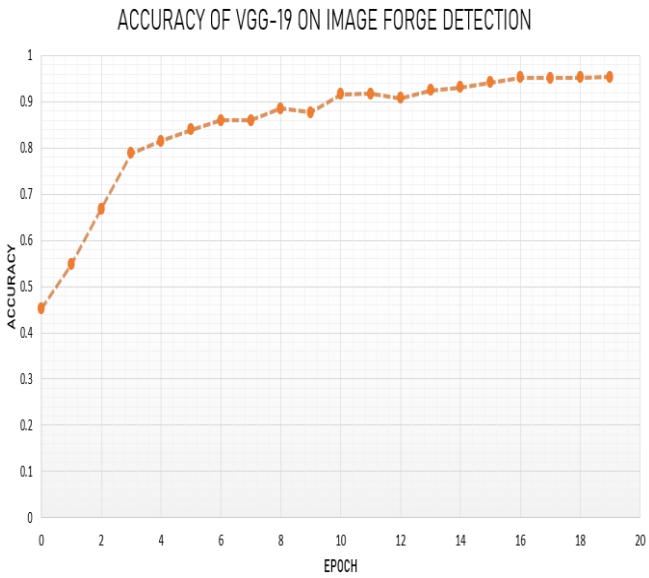

Figure 4. Accuracy of the DL model

Just like the training, the accuracy of the model is low during the initial epochs and rises rapidly as the number of epochs increases. By the final epoch of training, the accuracy of the model is greater than 95%. The loss values of the model are also recorded during testing and are shown in figure 5.

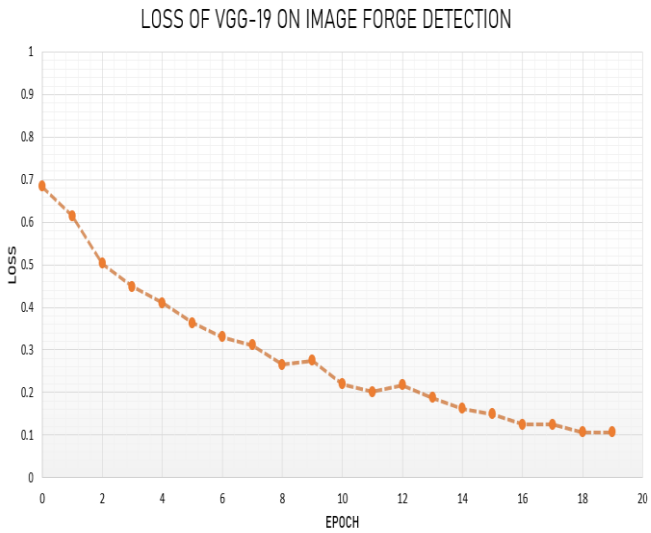

Figure 5. Loss of the DL model

The loss value is also too high during the first 8 epochs. But as the number of epochs increases, the loss value decreases and it reaches the value of 0.1 during the 18th and 19th epoch which is also great for a computer vision model.

VII. Conclusion

A dataset containing both original and forged images is gathered from the CASIA database. This dataset contains nearly 12000 images, with approximately 7500 original images and 5000 forged images. These images are then preprocessed to keep their dimensions consistent. Image resizing and denoising are examples of preprocessing techniques. Simultaneously, a transfer learning algorithm is used to create a DL model. The model is built using the VGG–19 architecture. The preprocessed images are then used to train the model. The training is repeated for several epochs, with the results recorded and tabulated for each one. As the training produced better results with the highest accuracy of 0.9535 and the lowest loss value of 0.1067, the model is then tested. Just like the training, the accuracy is higher than 0.95 and the loss value is lesser than 0.1 during the final epoch of testing. The results clearly explain the fact that this model is extremely efficient in detecting forged images. These days, it is so hard to predict false and forged images and videos because the fraudulent are so effective in concealing the major factors that can give away the fact that the media is forged. This issue can be resolved by the usage of the model which was developed in this study. This model can be deployed into a website or a software application making it useful for common people.

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